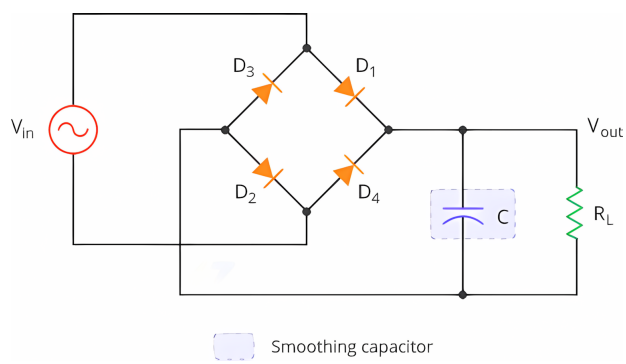
$PH3104\ Problem\ Set : II \quad (Total\ Marks : 15)$

Due on 10*th October* 2025

1. Starting from quasi Fermi level (see class notes) for n-type and p-type semiconductors in a p-n junction diode, derive the full diode equation, taking into account the drift as well as diffusion of carriers across the p-n junction. Explain all the terms. Give appropriate diagrams as well.

   [5]

2. For the circuit below, evaluate the Ripple factor for the output across the Load Resistor. [4]



3. Design an LC filter for a full-wave recifier (center tap) to provide an output voltage of 10 V with a load current of 200 mA. The ripple factor should be limited to 2% and the supply frequency is 50 Hz. Subsequently, calculate the Voltage Regulation of the circuit.

   [6]